# 11 WHAT EMERGES FROM A PARTICULARITY

*Why do the tall pine*
*and the pale poplar*
*intertwine their branches*
*to provide such sweet shade for us?*
*Why does the fleeting water*
*invent bright spirals*
*in the turbulent stream? (II, 9)*

## IT IS ENTROPY, NOT ENERGY, THAT DRIVES THE WORLD

At school I was told that it is energy that makes the world go round. We need to get energy—for example, from petroleum, from the sun, or from nuclear sources. Energy makes our engines run, helps plants to grow, and causes us to wake up every morning full of vitality.

But there is something that does not add up. Energy—as I was also told at school—is conserved. It is neither created nor destroyed. If it is conserved, why do we have to constantly resupply it? Why can't we just keep using the same energy?

The truth is that there is plenty of energy and it is not consumed. It's not energy that the world needs in order to keep going. What it needs is low entropy.

Energy (be it mechanical, chemical, electrical, or potential) transforms itself into thermal energy, that is to say, into heat: it goes into cold things, and there is no free way of getting it back from there to reuse it to make a plant grow, or to power a motor. In this process, the energy remains the same but the entropy increases, and it is *this* which cannot be turned back. The second law of thermodynamics demands it.

What makes the world go round are not sources of energy but sources of low entropy. Without low entropy, energy would dilute into uniform heat and the world would go to sleep in a state of thermal equilibrium—there would no longer be any distinction between past and future, and nothing would happen.